

trail-creek-exploratory

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Packages

```
library(tidyverse)
library(here)
library(janitor)
library(viridis)
library(wesanderson)
```

Data

```
compiled_isotope <- read_csv(here("data", "compiled.csv"), show_col_types =
FALSE) |>
  clean_names() |> # lowercase and underscores in column names
  rename(sample = sample_name, # renaming columns to be shorter
         delta_d = processed_delta_d,
         delta_d_std = processed_delta_d_stddev,
         delta_o = processed_delta_18o,
         delta_o_std = processed_delta_18o_stddev) |>
  mutate(date = mdy(date)) |> # telling r the date format
  filter(location %in% c("utc", "tc"))

site_order <- c("tc_isco", "tc_1", "tc_15", "tc_2", "tc_3",
               "utc_isco", "tc_ds", "tc_mid", "tc_us")

trail_creek <- compiled_isotope |>
  filter(location %in% c("utc", "tc")) |>
  mutate(site = factor(site, levels = site_order))

tc_isotope <- trail_creek |>
  filter(location %in% c("tc"))

utc_isotope <- trail_creek |>
  filter(location %in% c("utc"))

gradient_tc <- read_csv(here("data", "tc_gradient_data.csv"))
```

```

metadata <- read_csv(here("data", "metadata.csv"))

# Downstream Trail Creek
tc_data <- read_csv(here("data", "TrailCreek_2026-07-23-15.csv"),
show_col_types = FALSE) |> # reading in data
clean_names() |> # lowercase and underscores in column names
rename(date_time = date_time_mdt, # renaming columns to be shorter
dp = differential_pressure_k_pa,
ap = absolute_pressure_k_pa,
temp = temperature_c,
water_level = water_level_m,
bp = barometric_pressure_k_pa) |>
mutate(date_time_parsed = mdy_hms(date_time), # telling r the date format
date = as_date(date_time_parsed)) |> # making a date column
group_by(date) |>
summarise(avg_dp = mean(dp, na.rm = TRUE), # averaging the data based on
the day
avg_ap = mean(ap, na.rm = TRUE),
avg_temp = mean(temp, na.rm = TRUE),
avg_water_level = mean(water_level, na.rm = TRUE),
avg_bp = mean(bp, na.rm = TRUE),
total_readings = n(), # the amount of data it used to average
.groups = "drop") |>
mutate(site = "tc") # creating a new column of the site

# Upstream Trail Creek
utc_data <- read_csv(here("data", "UTC_2026-07-23-15.csv"), show_col_types =
FALSE) |> # reading in data
clean_names() |> # lowercase and underscores in column names
rename(date_time = date_time_mdt, # renaming columns to be shorter
dp = differential_pressure_k_pa,
ap = absolute_pressure_k_pa,
temp = temperature_c,
water_level = water_level_m,
bp = barometric_pressure_k_pa) |>
mutate(date_time_parsed = mdy_hms(date_time), # telling r the date format
date = as_date(date_time_parsed)) |> # making a date column
group_by(date) |>
summarise(avg_dp = mean(dp, na.rm = TRUE), # averaging the data based on
the day
avg_ap = mean(ap, na.rm = TRUE),
avg_temp = mean(temp, na.rm = TRUE),
avg_water_level = mean(water_level, na.rm = TRUE),

```

```

    avg_bp = mean(bp, na.rm = TRUE),
    total_readings = n(), # the amount of data it used to average
    .groups = "drop") |>
  mutate(site = "utc") # creating a new column of the site

tc_utc_data <- merge(tc_data, utc_data, all = TRUE) # merging the tc and utc data

isco_data <- merge(
  x = tc_utc_data,
  y = compiled_isotope,
  by.x = c("site", "date"), # columns in tc_utc_data
  by.y = c("location", "date") # matching columns in compiled_isotope
)

```

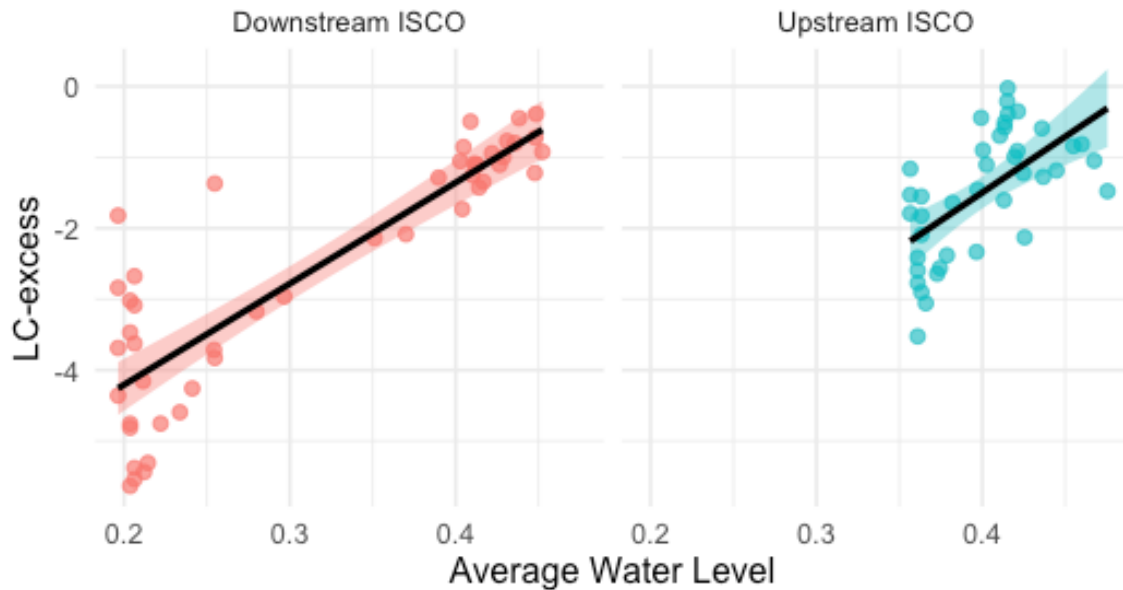
```

ggplot(data = isco_data, aes(x = avg_water_level, y = lc_excess, color = site)) +
  geom_point(size = 2, alpha = 0.7) +
  geom_smooth(method = "lm", aes(fill = site), color = "black", se = TRUE) +
  labs(
    title = "Relationship between Average Water Level and LC-excess",
    subtitle = "Analysis of isotope variations by downstream and upstream",
    x = "Average Water Level",
    y = "LC-excess"
  ) +
  theme_minimal(base_size = 12) +
  theme(plot.title = element_text(face = "bold", size = 14),
        legend.position = "none") +
  facet_wrap(~site, labeller = as_labeller(c("tc" = "Downstream ISCO",
                                             "utc" = "Upstream ISCO")))

```

Relationship between Average Water Level and LC-excess

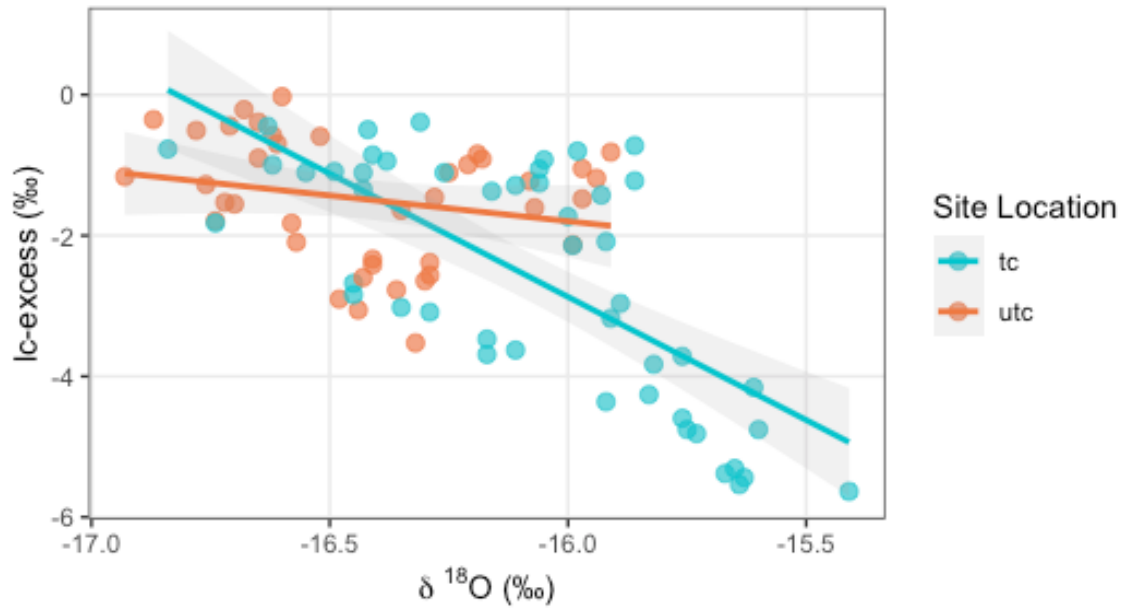
Analysis of isotope variations by downstream and upstream



```
ggplot(data = trail_creek, aes(x = delta_o, y = lc_excess, color = location)) +
  geom_point(size = 2.5, alpha = 0.7) +
  geom_smooth(method = "lm", se = TRUE, alpha = 0.15, size = 1) +
  scale_color_manual(values = c("tc" = "turquoise3", "utc" = "sienna2")) +
  labs(
    title = "Line-Conditioned Excess (lc-excess) vs. Delta 180",
    subtitle = "Comparing evaporation signals between downstream and upstream
sites",
    x = expression(paste(delta, " ^18, "0 (\u2030)")), # Formats as  $\delta^{18}O$  (‰)
    y = "lc-excess (\u2030)",
    color = "Site Location") +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
    legend.position = "right",
    panel.grid.minor = element_blank())
```

Line-Conditioned Excess (lc-excess) vs. Delta 18O

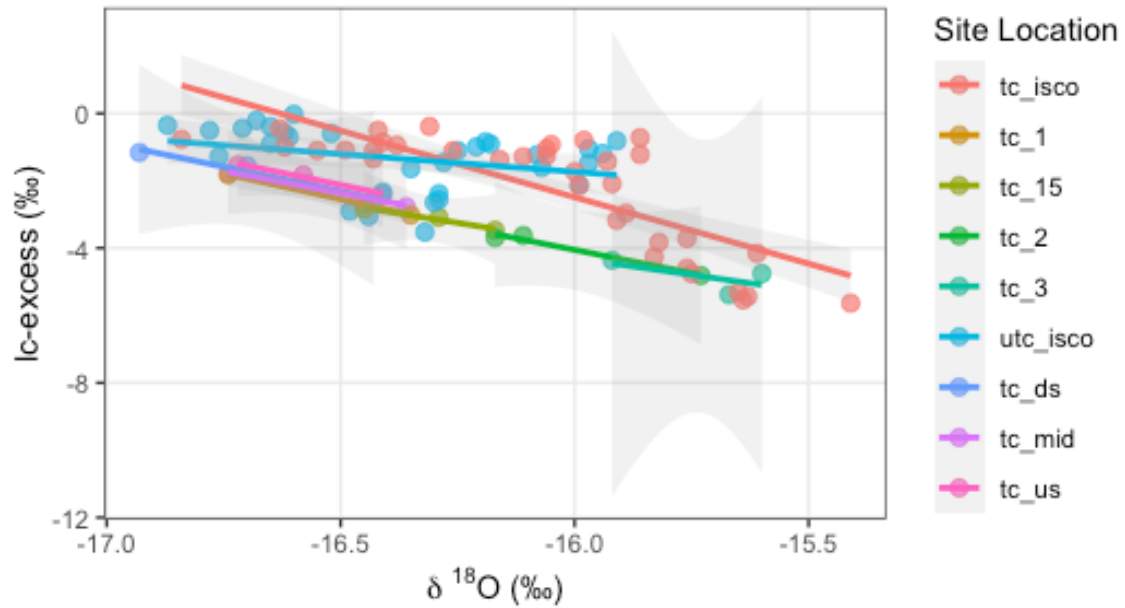
Comparing evaporation signals between downstream and upstream sites



```
ggplot(data = trail_creek, aes(x = delta_o, y = lc_excess, color = site)) +  
  geom_point(size = 2.5, alpha = 0.7) +  
  geom_smooth(method = "lm", se = TRUE, alpha = 0.15, size = 1) +  
  labs(  
    title = "Line-Conditioned Excess (lc-excess) vs. Delta 180",  
    subtitle = "Comparing evaporation signals across sites",  
    x = expression(paste(delta, " " ^18, "O (\u2030)")), # Formats as δ18O (‰)  
    y = "lc-excess (\u2030)",  
    color = "Site Location") +  
  theme_bw() +  
  theme(plot.title = element_text(face = "bold", size = 14),  
        legend.position = "right",  
        panel.grid.minor = element_blank())
```

Line-Conditioned Excess (lc-excess) vs. Delta 18O

Comparing evaporation signals across sites

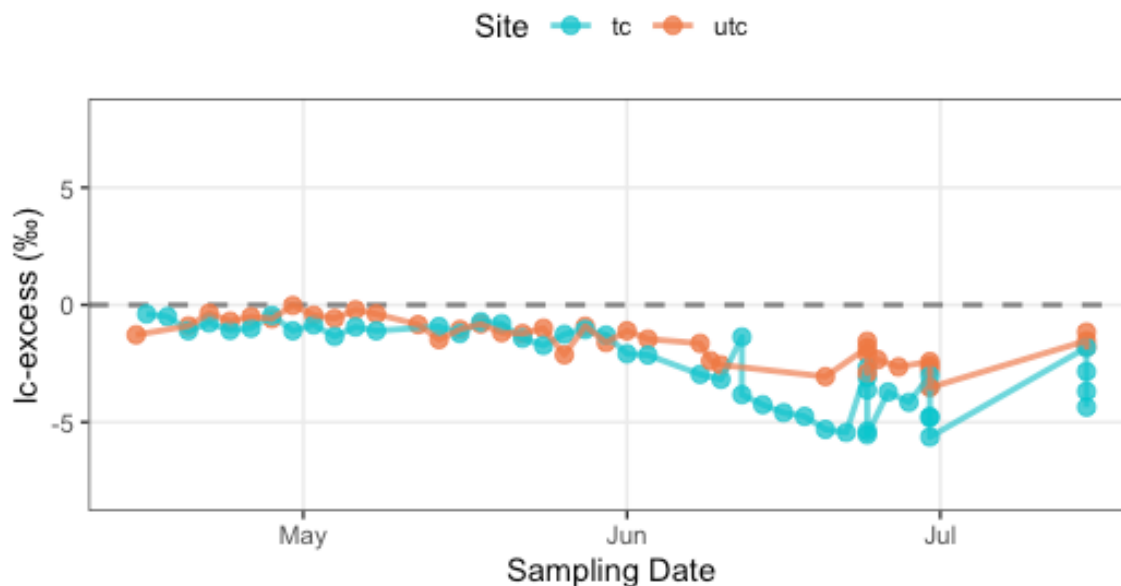


```
ggplot(data = trail_creek,
  aes(x = date,
    y = lc_excess,
    color = location,
    group = location)) +
  geom_hline(yintercept = 0,
    linetype = "dashed",
    color = "gray50",
    size = 0.8) +
  geom_point(size = 2.5,
    alpha = 0.8) +
  geom_line(size = 1,
    alpha = 0.7) +
  scale_color_manual(values = c("tc" = "turquoise3",
    "utc" = "sienna2")) +
  labs(title = "Time-Series of Line-Conditioned Excess (lc-excess)",
    subtitle = "Values below 0 indicate potential evaporative enrichment",
    x = "Sampling Date",
    y = "lc-excess (‰)",
    color = "Site") +
  ylim(-8,8) +
  theme_bw() +
```

```
theme(plot.title = element_text(face = "bold", size = 14),
      panel.grid.minor = element_blank(),
      legend.position = "top")
```

Time-Series of Line-Conditioned Excess (lc-excess)

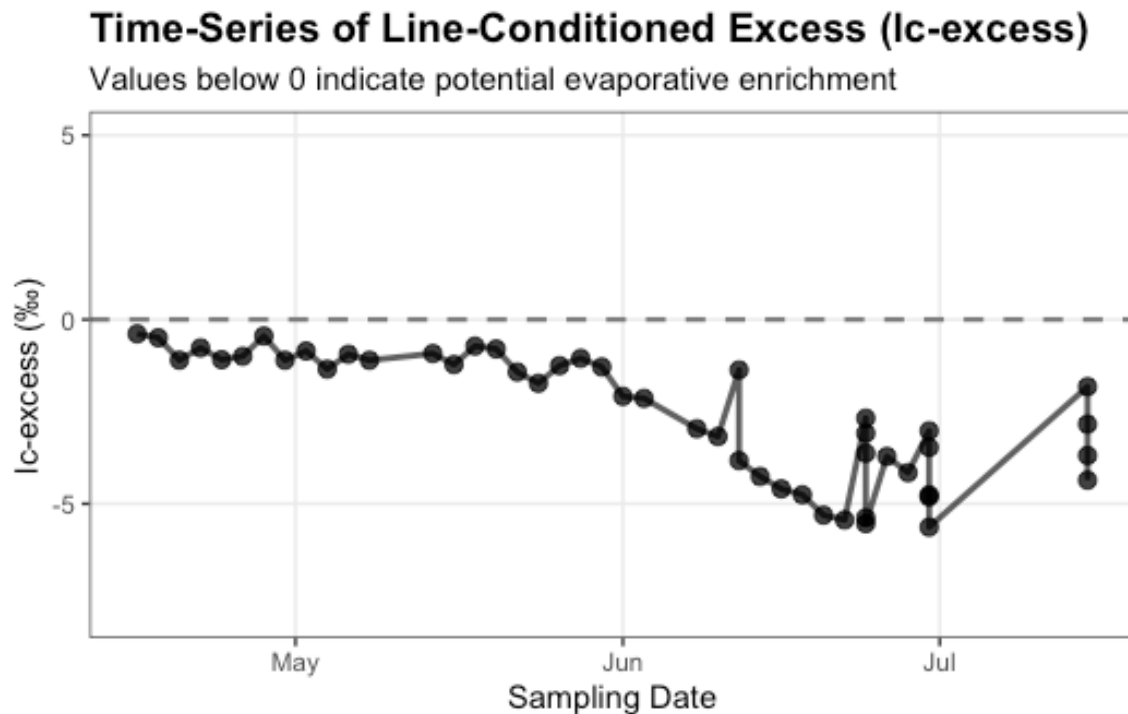
Values below 0 indicate potential evaporative enrichment



Time-Series of Line-Conditioned Excess (lc-excess)

```
ggplot(data = tc_isotope,
      aes(x = date,
          y = lc_excess)) +
  geom_hline(yintercept = 0,
            linetype = "dashed",
            color = "gray50",
            size = 0.8) +
  geom_point(size = 2.5,
            alpha = 0.8) +
  geom_line(size = 1,
            alpha = 0.7) +
  labs(title = "Time-Series of Line-Conditioned Excess (lc-excess)",
       subtitle = "Values below 0 indicate potential evaporative enrichment",
       x = "Sampling Date",
       y = "lc-excess (‰)",
       color = "Site") +
```

```
ylim(-8,5) +
theme_bw() +
theme(plot.title = element_text(face = "bold", size = 14),
      panel.grid.minor = element_blank(),
      legend.position = "top")
```



Time-Series of Line-Conditioned Excess (lc-excess)

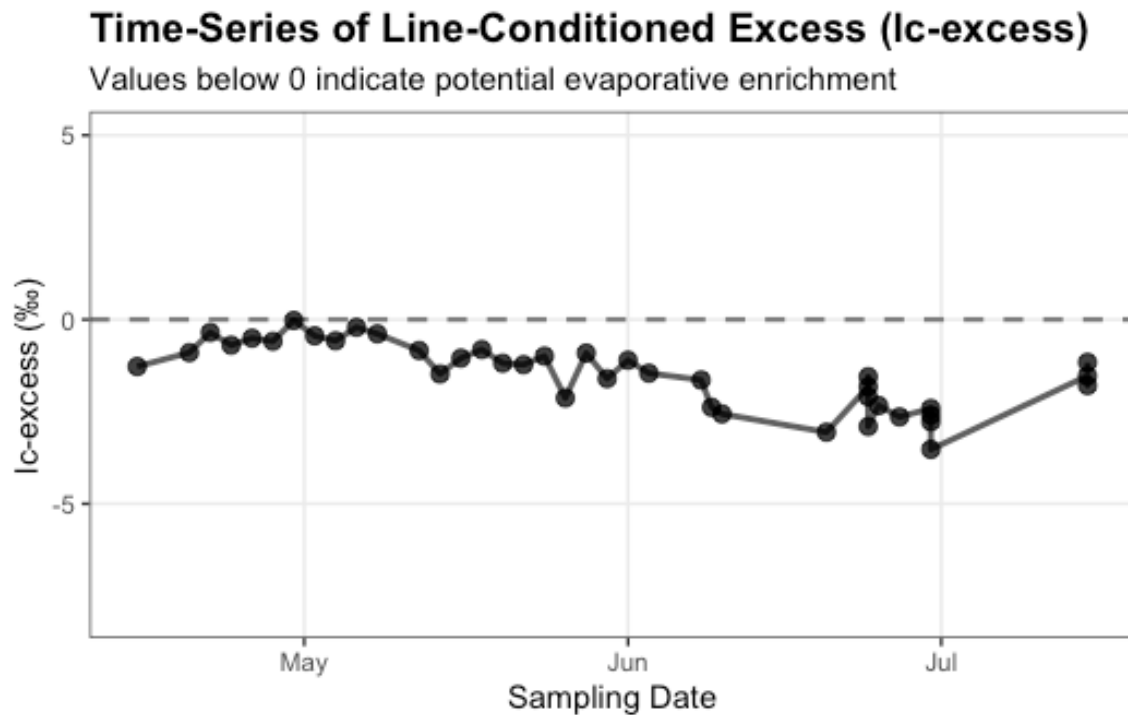
```
ggplot(data = utc_isotope,
       aes(x = date,
           y = lc_excess)) +
geom_hline(yintercept = 0,
           linetype = "dashed",
           color = "gray50",
           size = 0.8) +
geom_point(size = 2.5,
           alpha = 0.8) +
geom_line(size = 1,
           alpha = 0.7) +
labs(title = "Time-Series of Line-Conditioned Excess (lc-excess)",
     subtitle = "Values below 0 indicate potential evaporative enrichment",
     x = "Sampling Date",
```



```

    y = "lc-excess (\u2030)",
    color = "Site") +
ylim(-8,5) +
theme_bw() +
theme(plot.title = element_text(face = "bold", size = 14),
      panel.grid.minor = element_blank(),
      legend.position = "top")

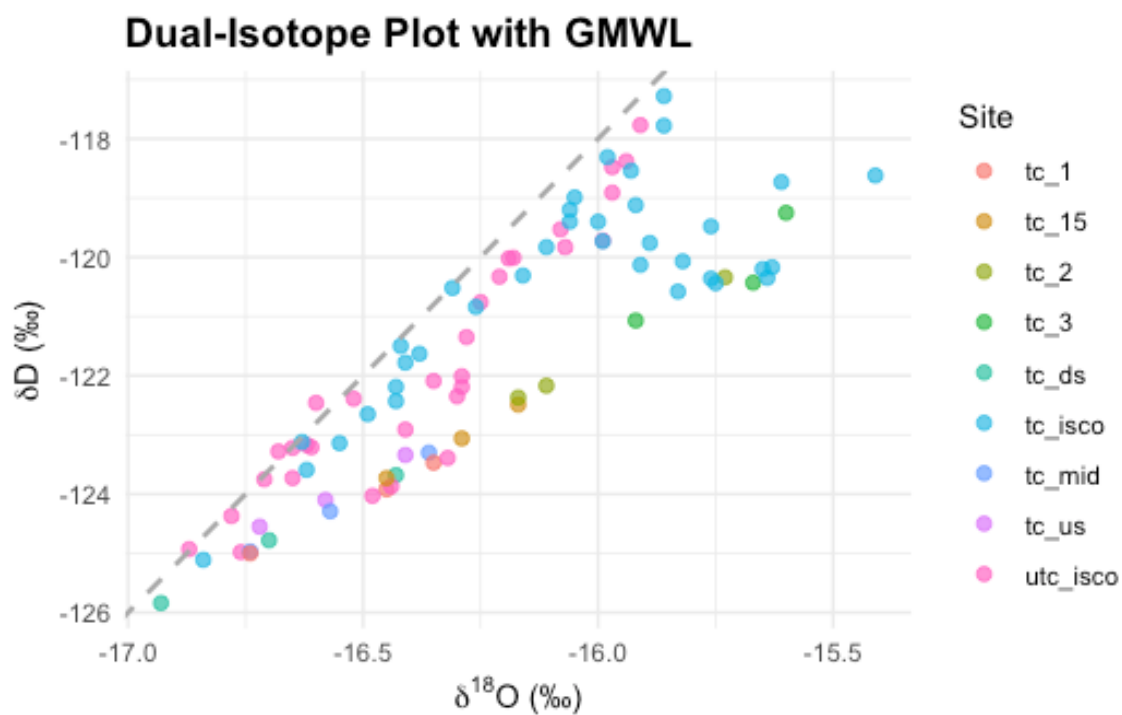
```



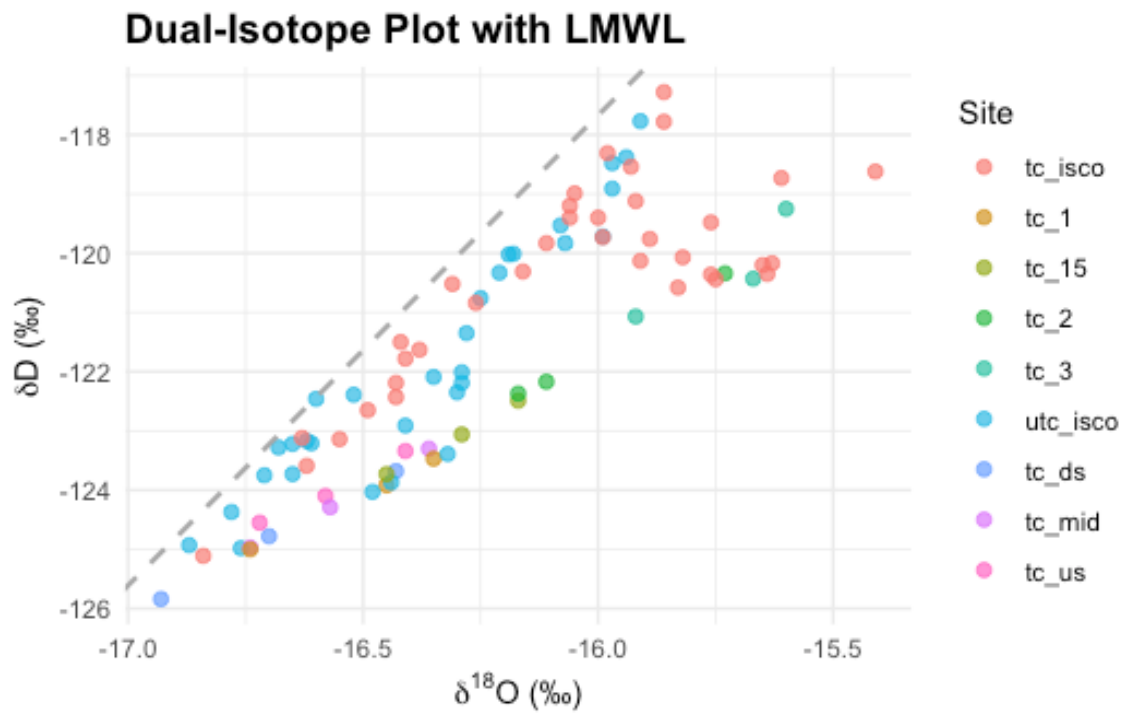
```

ggplot(compiled_isotope, aes(x = delta_o, y = delta_d, color = site)) +
  geom_point(alpha = 0.7, size = 2) +
  # Add Global Meteoric Water Line (GMWL: y = 8x + 10)
  geom_abline(intercept = 10, slope = 8, linetype = "dashed", color = "darkgray",
             linewidth = 0.8) +
  labs(
    title = "Dual-Isotope Plot with GMWL",
    x = expression(paste(delta^{18}, "O (‰)")),
    y = expression(paste(delta, "D (‰)")),
    color = "Site"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14))

```

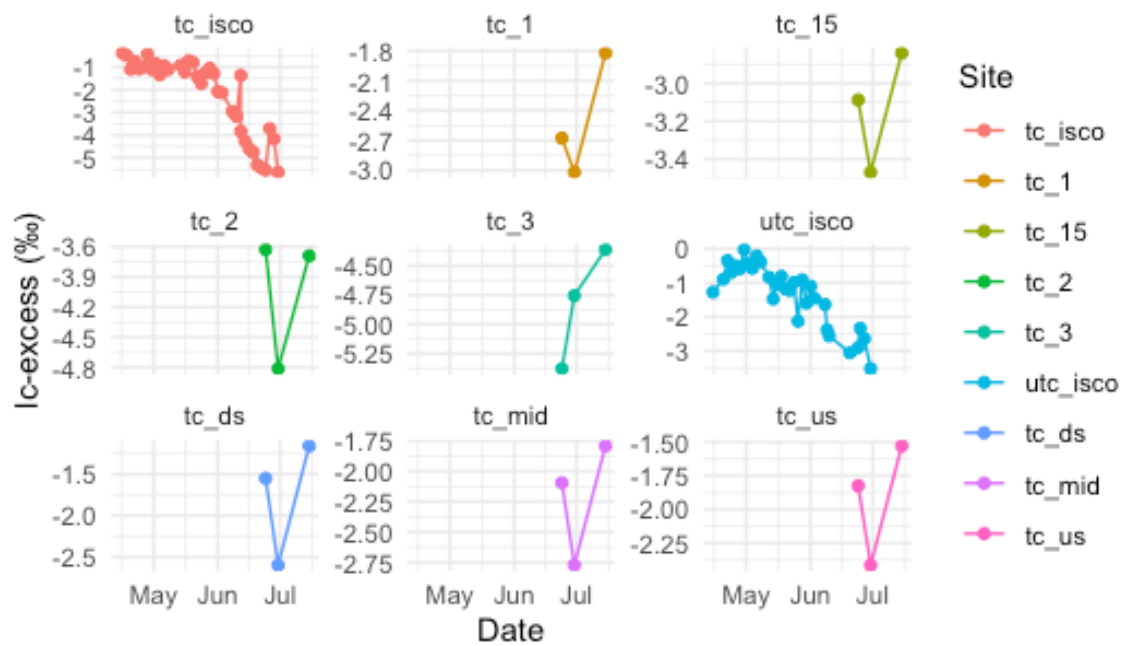


```
ggplot(trail_creek, aes(x = delta_o, y = delta_d, color = site)) +
  geom_point(alpha = 0.7, size = 2) +
  # Add Local Mean Water Line (LMWL: y = 7.94 x + 9.37 )
  geom_abline(intercept = 9.37, slope = 7.94, linetype = "dashed", color =
"darkgray", linewidth = 0.8) +
  labs(
    title = "Dual-Isotope Plot with LMWL",
    x = expression(paste(delta^{18}, "O (‰)")),
    y = expression(paste(delta, "D (‰)")),
    color = "Site"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14))
```

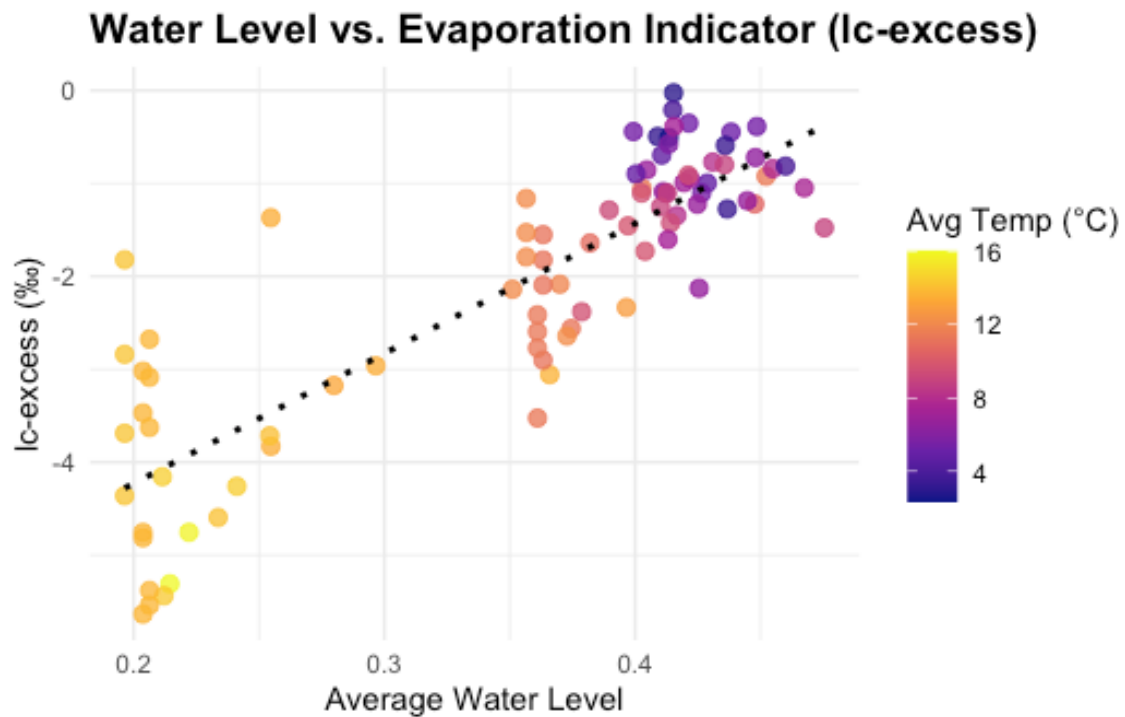


```
ggplot(trail_creek, aes(x = date, y = lc_excess, color = site)) +
  geom_line(linewidth = 0.5) +
  geom_point(size = 1.5) +
  labs(
    title = "Temporal Variations in Line-Conditioned Excess",
    x = "Date",
    y = "lc-excess (%)",
    color = "Site"
  ) +
  scale_fill_viridis_d(option = "B") +
  facet_wrap(~site, scales = "free_y") +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14))
```

Temporal Variations in Line-Conditioned Excess



```
ggplot(isco_data, aes(x = avg_water_level, y = lc_excess)) +
  geom_point(aes(color = avg_temp), size = 2.5, alpha = 0.8) +
  geom_smooth(method = "lm", color = "black", linetype = "dotted", se = FALSE) +
  scale_color_viridis_c(option = "plasma") + # Beautiful temperature gradient
  labs(
    title = "Water Level vs. Evaporation Indicator (lc-excess)",
    x = "Average Water Level",
    y = "lc-excess (%)",
    color = "Avg Temp (°C)"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14))
```

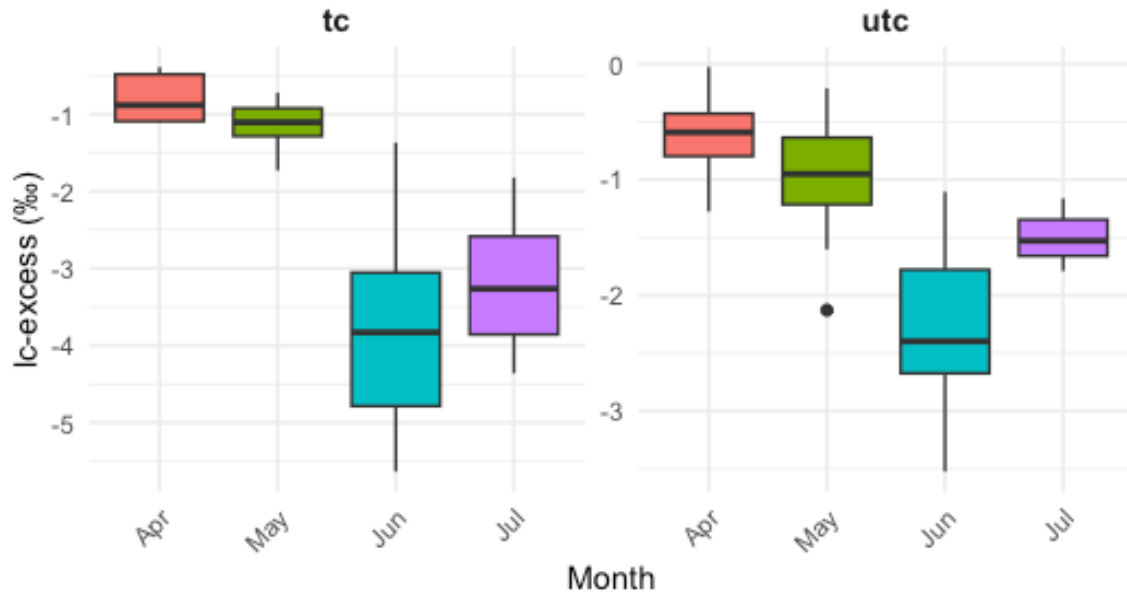


Monthly Variations in Evaporative Fraction (lc-excess) Across Study Sites

```
ggplot(trail_creek, aes(x = factor(month(date, label = TRUE, abbr = TRUE)), y =
lc_excess)) +
  geom_boxplot(aes(fill = factor(month(date))), show.legend = FALSE) +
  facet_wrap(~location, scales = "free_y") +
  labs(
    title = "Monthly Variations in Evaporative Fraction (lc-excess) Across
Study Sites",
    subtitle = "Seasonal isotopic shifts indicating changes in local evaporation-
condensation histories",
    x = "Month",
    y = "lc-excess (‰)"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14),
    strip.text = element_text(face = "bold", size = 11), # Emphasize site names
    axis.text.x = element_text(angle = 45, hjust = 1) # Tilt month labels if
they overlap
  )
```

Monthly Variations in Evaporative Fraction (lc-excess)

Seasonal isotopic shifts indicating changes in local evaporation-condensation



Dual-isotope plot ($\delta^{18}\text{O}$ vs δD) with LMWL reference

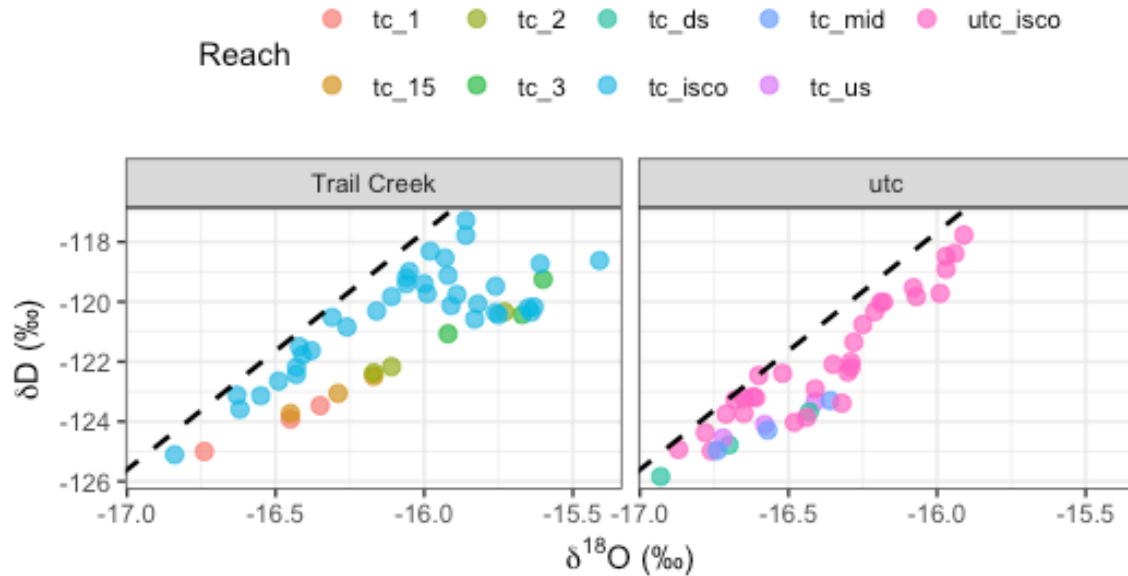
```
# Trail Creek LMWL calculated from OPIC

lmwl_slope <- 7.94
lmwl_intercept <- 9.37

ggplot(compiled_isotope, aes(x = delta_o, y = delta_d, color = site)) +
  geom_point(size = 2.5, alpha = 0.7) +
  geom_abline(slope = lmwl_slope, intercept = lmwl_intercept,
              linetype = "dashed", color = "black", linewidth = 0.8) +
  facet_wrap(~location, labeller = labeller(location = c(tc = "Trail Creek",
                                                         cc = "Coal Creek",
                                                         ic = "Italian
Creek")))) +
  labs(title = "Dual Isotope Plot:  $\delta^{18}\text{O}$  vs  $\delta\text{D}$ ",
       subtitle = "Dashed line = Trail Creek LMWL",
       x = expression(paste(delta^18, "O (‰)")),
       y = expression(paste(delta, "D (‰)")),
       color = "Reach") +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
        legend.position = "top")
```

Dual Isotope Plot: $\delta^{18}\text{O}$ vs δD

Dashed line = Trail Creek LMWL



Time series of lc-excess across season, colored by reach, faceted by watershed

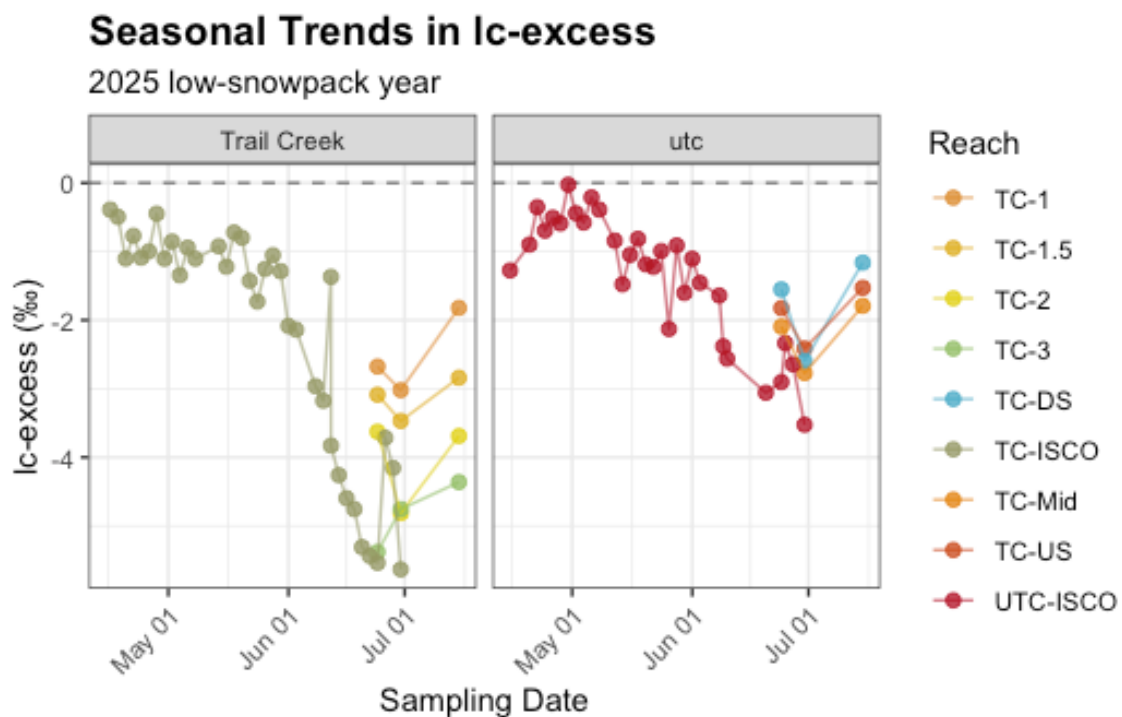
```
ggplot(compiled_isotope, aes(x = date, y = lc_excess, color = site, group =
interaction(site, location))) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray50") +
  geom_point(size = 2, alpha = 0.8) +
  geom_line(alpha = 0.6) +
  facet_wrap(~location, labeller = labeller(location = c(tc = "Trail Creek",
                                                         cc = "Coal Creek",
                                                         ic = "Italian Creek")))) +

  scale_x_date(date_labels = "%b %d") +
  scale_color_manual(values = wes_palette("FantasticFox1", n =
length(unique(compiled_isotope$site)),
                    type = "continuous"),
                    labels = c("coal_1" = "CC-1", "coal_2" = "CC-2", "coal_3"
= "CC-3", "coal_4" =
"CC-4", "coal_5" = "CC-5", "coal_6" = "CC-6",
"italian_1" = "IT-1",
"italian_2" = "IT-2", "italian_3" = "IT-3",
"italian_4" = "IT-4",
"italian_5" = "IT-5", "tc_isco" = "TC-ISCO",
"tc_1" = "TC-1", "tc_15"
= "TC-1.5", "tc_2" = "TC-2", "tc_3" = "TC-3",
```

```

"tc_ds" = "TC-DS",
                                "tc_mid" = "TC-Mid", "tc_us" = "TC-US", "utc_isco"
= "UTC-ISCO")) +
  labs(title = "Seasonal Trends in lc-excess",
        subtitle = "2025 low-snowpack year",
        x = "Sampling Date", y = "lc-excess (‰)", color = "Reach") +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
        legend.position = "right",
        axis.text.x = element_text(angle = 45, hjust = 1))

```



```

# averaging the lc-excess and including standard error
delta_summary <- compiled_isotope |>
  mutate(site = factor(site, levels = site_order)) |>
  group_by(site) |>
  summarise(
    mean_lc = mean(lc_excess, na.rm = TRUE),
    se_lc = sd(lc_excess, na.rm = TRUE) / sqrt(n()),
    .groups = "drop"
  )

# creating a plot to show the mean lc-excess of each site

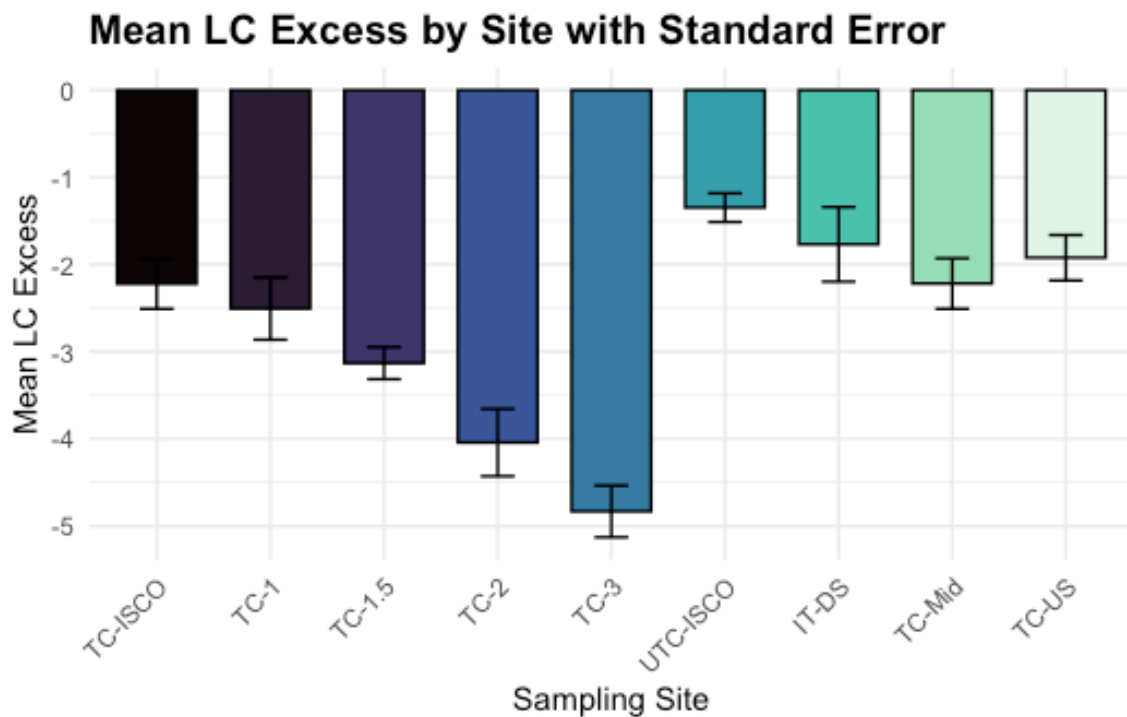
```



```

ggplot(delta_summary, aes(x = site, y = mean_lc, fill = site)) +
  geom_col(color = "black", width = 0.7) +
  geom_errorbar(aes(ymin = mean_lc - se_lc, ymax = mean_lc + se_lc),
    width = 0.3,
    color = "black") +
  scale_x_discrete(limits = c("tc_isco", "tc_1", "tc_15", "tc_2", "tc_3",
    "utc_isco", "tc_ds", "tc_mid", "tc_us"),
    labels = c("tc_isco" = "TC-ISCO", "tc_1" = "TC-1", "tc_15" = "TC-1.5", "tc_2"
    = "TC-2",
      "tc_3" = "TC-3", "tc_ds" = "IT-DS", "tc_mid" = "TC-Mid", "tc_us" =
      "TC-US",
      "utc_isco" = "UTC-ISCO")) +
  scale_fill_viridis_d(option = "G") + # color palette
  theme_minimal() +
  labs(title = "Mean LC Excess by Site with Standard Error",
    x = "Sampling Site",
    y = "Mean LC Excess") +
  theme(plot.title = element_text(face = "bold", size = 14),
    axis.text.x = element_text(angle = 45, hjust = 1), # tilt labels
    legend.position = "none")

```



```

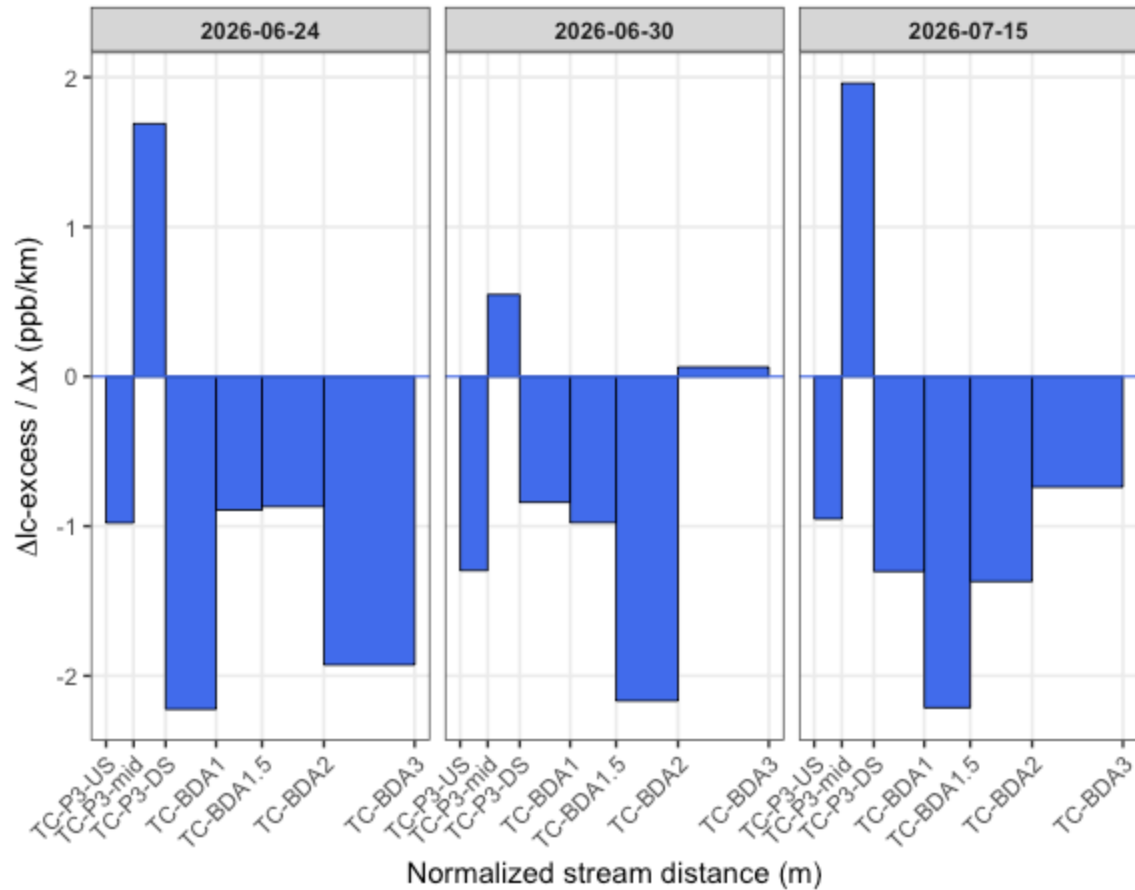
# Building a look up table of stream distances
distances <- tibble(site = c("TC-P3-US", "TC-P3-mid", "TC-P3-DS", "TC-BDA1",
                             "TC-BDA1.5", "TC-BDA2", "TC-BDA3"),
                    dist_m = c(57.6, 333, 654, 1160, 1620, 2240, 3150)) |>
  mutate(xmax = dist_m,
         xmin = lag(dist_m, default = 0)) # previous site's distance, 0 for first

# Creating a filtered data frame for lc-excess
gradient_lc_clean <- gradient_tc |>
  select(site, normalized_stream_distance_m, `Δlc-excess/Δx_6-24`, `Δlc-excess/
Δx_6-30`, `Δlc-excess/Δx_7-15`) |>
  pivot_longer( cols = c(`Δlc-excess/Δx_6-24`, `Δlc-excess/Δx_6-30`, `Δlc-
excess/Δx_7-15`),
               names_to = "date",
               values_to = "gradient") |>
  mutate(date = case_when(date == "Δlc-excess/Δx_6-24" ~ "2026-06-24",
                          date == "Δlc-excess/Δx_6-30" ~ "2026-06-30",
                          date == "Δlc-excess/Δx_7-15" ~ "2026-07-15"),
         gradient = as.numeric(gradient)) |>
  left_join(distances %>% select(site, xmin, xmax), by = "site") |>
  mutate(site = factor(site, levels = distances$site)) |>
  filter(!is.na(gradient))

# Gradient plot
ggplot(gradient_lc_clean) +
  geom_rect(aes(xmin = xmin, xmax = xmax, ymin = 0, ymax = gradient),
           fill = "royalblue2", color = "black", linewidth = 0.3) +
  geom_hline(yintercept = 0, color = "royalblue2", linewidth = 0.4) +
  facet_wrap(~ date, nrow = 1) +
  scale_x_continuous(
    breaks = distances$dist_m,
    labels = distances$site) +
  labs(title = "Trail Creek: Spatial variations in Δlc-excess/Δx across sampling
dates",
       x = "Normalized stream distance (m)",
       y = expression(paste(Delta, "lc-excess / ", Delta, "x (ppb/km)"))) +
  theme_bw(base_size = 11) +
  theme(plot.title = element_text(face = "bold", size = 14),
        panel.grid.minor = element_blank(),
        strip.text = element_text(face = "bold"),
        axis.text.x = element_text(angle = 45, hjust = 1))

```

Trail Creek: Spatial variations in $\Delta\text{lc-excess}/\Delta x$ across samp



```
gradient_cum_tc <- gradient_tc |>
  select(site, cumulative_dams,
         `Δlc-excess/Δcum_dam_length_6-24`, `Δlc-excess/Δcum_dam_length_6-30`,
         `Δlc-excess/Δcum_dam_length_7-15`) |>
  pivot_longer(
    cols = c(`Δlc-excess/Δcum_dam_length_6-24`, `Δlc-excess/Δcum_dam_length_6-30`, `Δlc-excess/Δcum_dam_length_7-15`),
    names_to = "date",
    values_to = "gradient"
  ) |>
  mutate(
    date = case_when(
      date == "Δlc-excess/Δcum_dam_length_6-24" ~ "2026-06-24",
      date == "Δlc-excess/Δcum_dam_length_6-30" ~ "2026-06-30",
      date == "Δlc-excess/Δcum_dam_length_7-15" ~ "2026-07-15"
    ),
  )
```

```

    gradient = as.numeric(gradient),
    cumulative_dams = as.numeric(cumulative_dams),
    site = factor(site, levels = c("TC-P3-US", "TC-P3-mid", "TC-P3-DS",
                                   "TC-BDA1", "TC-BDA1.5", "TC-BDA2", "TC-BDA3"))
  ) |>
  filter(!is.na(gradient))

ggplot(gradient_cum_tc, aes(x = site, y = gradient, fill = site)) +
  geom_col(color = "black", width = 0.7) +
  geom_hline(yintercept = 0, color = "steelblue", linewidth = 0.4) +
  facet_wrap(~ date, nrow = 1) +
  scale_fill_viridis_d(option = "G") +
  scale_x_discrete(labels = function(x) {
    lookup <- setNames(gradient_cum_tc$cumulative_dams, gradient_cum_tc$site)
    lookup[x]
  }) +
  labs(
    x = "Cumulative dam count",
    y = expression(Delta * "lc-excess/" * Delta * "cumulative dam length (" *
"\u2030" * ")/m")
  ) +
  theme_bw(base_size = 12) +
  theme(
    strip.text = element_text(face = "bold", size = 11),
    strip.background = element_rect(fill = "gray85", color = NA),
    panel.grid.minor = element_blank(),
    panel.grid.major.x = element_blank(),
    legend.title = element_text(face = "bold")
  )

```

